

Ethanol Blended Fuels

A live note from a candidate’s notebook. Built from five articles over several months, and still one page.

PAPER	GS 3 › Infrastructure › Energy › Oil and Natural Gas
ALSO USEFUL FOR	GS 3 Economy (energy security) · GS 3 Environment (biofuels) · GS 3 Agriculture
PRIORITY	High
LAST UPDATED	17–08–2026
BUILT FROM	Five articles

NEW SINCE 17-08-2026

If you last revised this note before August, these five sections are the only things you have not seen. Nothing else changed; nothing was removed.

Vehicle compatibility — the older fleet

Green water vs Blue water

Engine-damage claims — evidence check

Framing principle — reversibility

Import-bill economics

HOW TO READ THIS NOTE

[+17-08-2026]	Printed in the margin beside the line it belongs to. When a topic returns, skim only the dated lines — you never re-read a note you have already revised.
Trap	A likely Prelims trap: a reversed definition, the wrong ministry, a false “first”.
(revision)	Static material kept in place for revision, not new in this pass.

ETHANOL BLENDING PROGRAMME (revision)

- **Aim:** blend ethanol with petrol to reduce fossil-fuel dependence, lower crude-oil imports and promote cleaner mobility.
- **E20 fuel:** petrol containing 20% ethanol and 80% petrol. India achieved 20% blending in **2025**, ahead of the original 2030 target.
- Blending rose from about **1.5% in 2013–14** to nearly **20% by 2025**. Annual consumption under the programme is approaching **700 crore litres**.
- **Timeline:** **2003** initial blending programme · **2022** E10 achieved · **2025** E20 achieved · **2026** target, nationwide E20 availability.

ETHANOL SOURCES

- **Sugar-based:** sugarcane juice, molasses. **Grain-based:** maize, damaged food grains, surplus rice stocks.

Trap
[+17-08-2026]

- **Maize currently supplies about half of India's ethanol; grains together supply nearly 67% of total feedstock** — competing for the same land as soybean, groundnut and mustard.

SECOND-GENERATION ETHANOL *(revision)*

- Produced from lignocellulosic biomass — agricultural residue and non-food biomass.

Trap

- 2G ethanol uses lignocellulosic biomass, **not** primarily edible grains.

FLEX-FUEL VEHICLES

- Can operate on petrol–ethanol blends with higher ethanol content; require engine modifications for high-ethanol compatibility.

VEHICLE COMPATIBILITY — THE OLDER FLEET

new · 17-08-2026

Trap
[+17-08-2026]

- India has roughly **75 to 80 million older two-wheelers** (built before BS4) with carburettor engines, which cannot sense oxygen content in the fuel blend. On E20 these engines draw in too little fuel for the air intake and run hot. Older rubber seals not rated for ethanol also degrade on contact with the fuel, regardless of engine temperature.

[+17-08-2026]

- The original ethanol roadmap had anticipated this and called for a lower-blend fuel to remain on sale for these vehicles — that option quietly vanished from pumps.

[+17-08-2026]

- **CEA recommendation:** restore **E10** at pumps for the pre-BS4 fleet, alongside continued E20 availability, while retrofit programmes catch up.

ENGINE-DAMAGE CLAIMS — EVIDENCE CHECK

new · 17-08-2026

Trap
[+17-08-2026]

- Viral claims that E20 “wrecks engines” are not supported by evidence. Ethanol carries about **two-thirds** the energy content of petrol, so the penalty at E20 is only **6-7%**, not the ~30% claimed online. Carbon monoxide and unburnt hydrocarbon emissions actually **fall** at E20.

Trap
[+17-08-2026]

- Bodies confirming no increased engine wear at E20 for ethanol-rated vehicles: **Oak Ridge National Laboratory** (US) — 86 vehicles, ~10 million km cumulative on blends up to E20; **ARAI**; **Indian Oil**.

- [+17-08-2026]
- Moving beyond E20 (to E27 or E30) is “a different decision altogether” — that is the threshold at which fuel begins to meaningfully crowd out food production.

RICE, BUFFER STOCKS AND THE FOOD-FUEL QUESTION *(revision)*

- India rice production (2025–26): **~151 MMT**. India is the **second-largest producer** after China and the **largest exporter** — exports (2024–25) **~24.5 MMT**, about **40% of global rice trade**.
- **FCI** procures food grains at Minimum Support Price and maintains buffer stocks for food security. Surplus FCI rice supplied for ethanol production is what raises the food-versus-fuel debate.

IMPORT-BILL ECONOMICS

new · 17-08-2026

- Trap

[+17-08-2026]
- Crude oil import bill: **~₹11–12 lakh crore a year**. Edible oil import bill: **~₹1.6–1.75 lakh crore a year** — the edible-oil gap is comparatively smaller and more closeable.
- [+17-08-2026]
- **Ethanol blending at E20 trims only 3–4% of the crude oil import bill.**
- [+17-08-2026]
- India currently produces **~40%** of its cooking-oil requirement domestically and aims to reach **72% by 2030–31**.

ENVIRONMENTAL FACTS

- **EROI** (energy return on investment): sugarcane ethanol $\approx$ **2–4**; grain-based ethanol $\approx$ **1.2–2**.
- **Energy content:** petrol $\approx$ 32 MJ/litre; ethanol $\approx$ 21 MJ/litre — E20 therefore carries **~6–7%** less energy than pure petrol.
- Sugarcane needs **~1,500–2,500 mm** of water across the growing season; sugarcane-ethanol’s water footprint may reach **~2,000–3,500 litres of water per litre of ethanol**.
- High EROI does **not** automatically mean low environmental impact. Energy efficiency and water footprint are separate parameters.
- Water-intensive sugarcane cultivation is significant in **Maharashtra** and **Karnataka**, where groundwater stress makes additional irrigation demand problematic.

GREEN WATER VS BLUE WATER

new · 17-08-2026

Read this against the 2,000–3,500 litre figure above. That number was not replaced — this is how to use it in an answer.

Trap
[+17-08-2026]

- **Green water** is rainfall the crop would have received anyway — not an additional cost. **Blue water** is drawn from rivers, canals, groundwater pumps and wells; it is Blue water that is genuinely scarce. Commonly cited “total water use” figures mix the two and **overstate the footprint**. Water-scarcity policy should target Blue-water use specifically, not the frightening combined total.

Trap
[+17-08-2026]

- **The climate case for ethanol beyond E20 is unsettled.** Any blending past E20 would come almost entirely from grain, and Indian life-cycle studies do not agree on whether grain ethanol is cleaner once cultivation and processing are counted.

BRAZIL ETHANOL MODEL (revision)

- Blending began **1931**; the **National Alcohol Programme (Proálcool)** launched **1975** after the 1973 oil crisis.
- **Flex-fuel vehicles** introduced commercially in 2003 — petrol, ethanol or any blend, with consumers choosing by price and availability. Brazil integrated fuel policy, automobile technology and consumer choice around one feedstock: sugarcane.

WHY ETHANOL CAN BE BOTH GREEN AND PROBLEMATIC

BENEFIT Ethanol blending ↑ → crude requirement ↓ → oil imports ↓ → energy security ↑

BUT Sugarcane cultivation → irrigation demand ↑ → groundwater extraction ↑ → water stress ↑

THEREFORE A low-carbon fuel is not necessarily a low-water-footprint fuel.

The correct question is not “how much ethanol can India blend?” but “how much ethanol can India sustainably produce, from which feedstock, and in which region?”

FRAMING PRINCIPLE — REVERSIBILITY

new · 17-08-2026

[+17-08-2026]

- Some policy levers adjust easily mid-course — procurement price, protection tariffs, water rules, edible-oil import duty. Others become structurally **locked in** once land, water, distillery capacity and cropping patterns are built around them: the blend mandate itself. Prudence dictates correcting the reversible levers first and holding the irreversible one until a thorough cost–benefit analysis is done.

[+17-08-2026]

- But holding at E20 also carries costs — distillery capacity built for more, loans taken against expected demand, and a sugar surplus that ethanol helps absorb. Both sides must be weighed.

MAINS POINTS *GS 3 — Environment, Energy, Agriculture*

Cleaner mobility pathway	Reduces fossil-fuel dependence and the crude import burden. <i>India achieved E20 in 2025, ahead of the 2030 timeline.</i>
Energy security — import substitution	Domestic ethanol production reduces vulnerability to global oil shocks. <i>A complement to the EV transition — though E20 trims only 3–4% of the crude import bill. [+17–08–2026]</i>
Rural economy — farmer income support	Creates additional demand for agricultural produce and residues. <i>Cane farmers' dues cleared and incomes lifted through the ethanol industry. [+17–08–2026]</i>
Food vs fuel — resource allocation conflict	Diverting edible grains towards fuel conflicts with food-security goals. <i>Maize now competes directly with soybean, groundnut and mustard for land, while distillation by-product feed undercuts soybean-meal prices — hurting oilseed farmers twice. [+17–08–2026]</i>
Agricultural distortion — MSP-induced cropping pattern	Assured procurement encourages water-intensive crops despite ecological stress. <i>Paddy cultivation in Punjab and Haryana → groundwater depletion.</i>
Water–energy–food nexus	Biofuel expansion must account for water, emissions and land pressure — but water figures must separate Green from Blue water, since only Blue water is genuinely scarce. [+17–08–2026] <i>Cane ethanol's reliance on already-stressed Blue water in Maharashtra and Karnataka.</i>
Policy design — the reversibility principle	Instruments that reverse easily should be corrected first; near-irreversible commitments held back until fully costed. [+17–08–2026] <i>Fix price, water and import-duty distortions now; hold the blend at E20 rather than escalating to E27 or E30.</i>
Way forward	Expand 2G ethanol and agricultural-waste utilisation · diversify feedstock · encourage less water-intensive crops · rationalise MSP incentives · promote flex-fuel vehicles · restore a lower-blend E10 option for the older fleet while retrofits scale [+17–08–2026] · target water rules at Blue-water use, not combined totals [+17–08–2026]

STATIC LINKS

National Policy on Biofuels, 2018 · Ethanol Blended Petrol Programme · 1G vs 2G biofuels · Energy security · Food–fuel–water nexus · Life-cycle assessment

PRACTICE QUESTIONS

PRELIMS — 1

With reference to ethanol blending in India, consider the following statements:

1. Ethanol can be produced from both sugar-based and grain-based feedstocks.
2. E20 fuel contains 20% petrol and 80% ethanol.
3. Ethanol blending can help reduce crude-oil import dependence.

How many are correct? **(b) Only two.** Statement 2 is reversed — E20 is 20% ethanol.

PRELIMS — 2 [+17-08-2026]

With reference to India's ethanol-blending programme, consider the following statements:

1. India's older pre-BS4 two-wheeler fleet with carburettor engines faces genuine compatibility concerns with E20.
2. Oak Ridge National Laboratory, ARAI and Indian Oil found no increased engine wear at E20 for ethanol-rated vehicles.
3. "Green water" refers to water extracted from rivers, canals and groundwater specifically for crop irrigation.

How many are correct? **(b) Only two.** Statement 3 describes **Blue** water, not Green.

MAINS — 250 WORDS

"India's ethanol-blending programme represents an important energy-security strategy, but its sustainability depends on the feedstock and geography of production." Discuss.

ALSO USE THIS NOTE FOR

[Cross-link](#)

- **GS 3 › Agriculture › Major Crops, GI Crops** — the feedstock shift's impact on oilseed cultivation. Maize is pulling land away from soybean, groundnut and mustard, and distillation by-product feed is undercutting soybean-meal prices.

One link. Ethanol touches five other pages in this notebook — fertilisers, food security, electric vehicles, compressed biogas, climate change — but only one of them is somewhere a real Mains answer would actually take you. Cross-links stay few enough to be worth following.

WHERE THIS NOTE CAME FROM

17-08-2026	<i>The Indian Express</i> , p.10 — "Bring back E10 for older vehicles. Work out food versus fuel trade-off" V. Anantha Nageswaran, Chief Economic Advisor
Earlier	EROI, water footprint and life-cycle sustainability
Earlier	Brazil's Proálcool model and flex-fuel adoption
Earlier	Rice, FCI buffer stocks and the food-versus-fuel question
Earlier	Ethanol Blending Programme fundamentals — E20, feedstocks, timeline

Five articles arrived on this topic over several months. Each one was placed into the page that already existed, in the section where it belonged, carrying the date it arrived. Nothing was written twice and nothing was deleted.

FIVE ARTICLES · ONE PAGE · NOTHING NOTED TWICE